

SPAR Friday meeting — 15 May 2026

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Headline

A prior issue was preventing the consciousness root from materially affecting the model's expert-rating predictions. A targeted fix moves things in the right direction but is incremental: even after it, the prior-predictive at the leaf level isn't where it should be, and middle-spectrum systems (Chicken, current LLMs) still aren't well-identified.

Where this week landed

The plan was synthetic validation: generate data from the model with a known truth, fit, see whether we recover it. Toy and intermediate setups recover correctly. The full baseline didn't, and on inspection the issues turned out to be upstream of the validation. Two prior / structural problems are blocking the model from doing what it should. So this talk covers:

- **Issue 1:** the prior wasn't transmitting consciousness to the ratings. Targeted fix in place; recovery looks reasonable on a 10-seed pilot, but the prior-predictive is still worse than I'd want.
- **Issue 2:** even after Issue 1, the binary root + Human / ELIZA-only anchor design can't well-identify C for systems in the middle of the spectrum.

Synthetic validation results exist on the handover branch, but they're more meaningful once Issues 1 and 2 are addressed. Synthetic data drawn from a misspecified prior doesn't tell us much about real-data behaviour, and the synthetic dataset itself doesn't resemble the real expert ratings for middle systems (more on that under Issue 2).

Notation. $R \in \{0, 1\}$ is the binary root state, the realised “is this system conscious?” outcome at a single draw. $C = P(R = 1)$ is the probability of consciousness, the per-system parameter we're trying to estimate.

Issue 1 — the prior wasn't transmitting consciousness to the ratings

What's wrong (intuitively)

Even if the model treats Human as conscious ($R = 1$) and ELIZA as not ($R = 0$), under the original priors that signal got diluted travelling down the indicator tree. By the time it reached the leaves, the predicted rating distributions for Human and ELIZA were almost identical — so the root barely changed what experts were predicted to write. In effect, the model said “consciousness doesn't really matter for the ratings.”

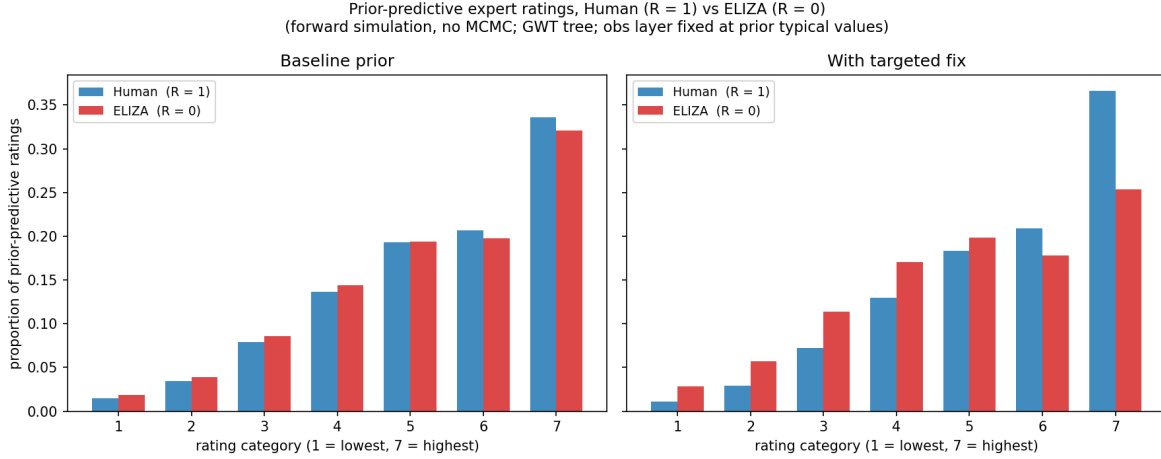


Figure 1: Prior-predictive expert ratings, Human vs ELIZA, baseline prior vs targeted fix

Visual — prior-predictive expert rating distributions

Forward simulation only (no MCMC) on the GWT tree, with the observation layer fixed at typical prior values so the q_j -driven story is visible. Two systems overlaid per panel: Human ($R = 1$, blue), ELIZA ($R = 0$, red).

Read off:

- **Baseline prior (left).** Human and ELIZA bars are nearly indistinguishable at every category. The model’s prior says essentially the same thing about expert ratings whether the system is conscious or not — root signal isn’t reaching the leaves.
- **With the targeted fix (right).** Visible separation appears: Human concentrates more at cat 7 (~0.36) while ELIZA shifts down (cat 7 ~0.25, more mass at cats 2–4).

Honest framing of where this leaves us. The fix is an improvement in direction but is still incremental. Even in the right panel, the ELIZA distribution doesn’t concentrate near the lowest ratings — both systems still lean noticeably high, and they’re more similar to each other than I’d want. So this isn’t “we’ve fixed the prior” — it’s “we’ve moved from a prior that essentially ignored consciousness, to one that registers it weakly.” A more principled revision (see follow-up) is the natural next step.

Mean prior-implied q_j across the 50 GWT indicators (sanity check):

Variant	Human ($R = 1$)	ELIZA ($R = 0$)	Gap
Baseline	0.638	0.591	0.047
Targeted fix	0.687	0.437	0.250

Drill-down on a single indicator — Stable Personality

To make the same story concrete on a leaf that domain experts will recognise, here's the same forward simulation zoomed in on one specific indicator. **Stable Personality** under GWT, depth 3, with path:

GWT → Coherence → Point of View → Stable Personality

Two of this indicator's three path edges (Point of View, Stable Personality) sit inside the targeted-fix override set; the top edge (Coherence) does not. So the baseline-vs-fix contrast on this leaf should reflect the override fairly directly.

For each prior draw we sample $(\beta^{\text{pres}}, \beta^{\text{abs}})$ at every edge along this path from its logit-Normal prior, propagate q_j down the path, and compute the implied rating distribution under typical observation params. 100 prior draws per variant per root state, faint coloured curves; medians overlaid in solid.

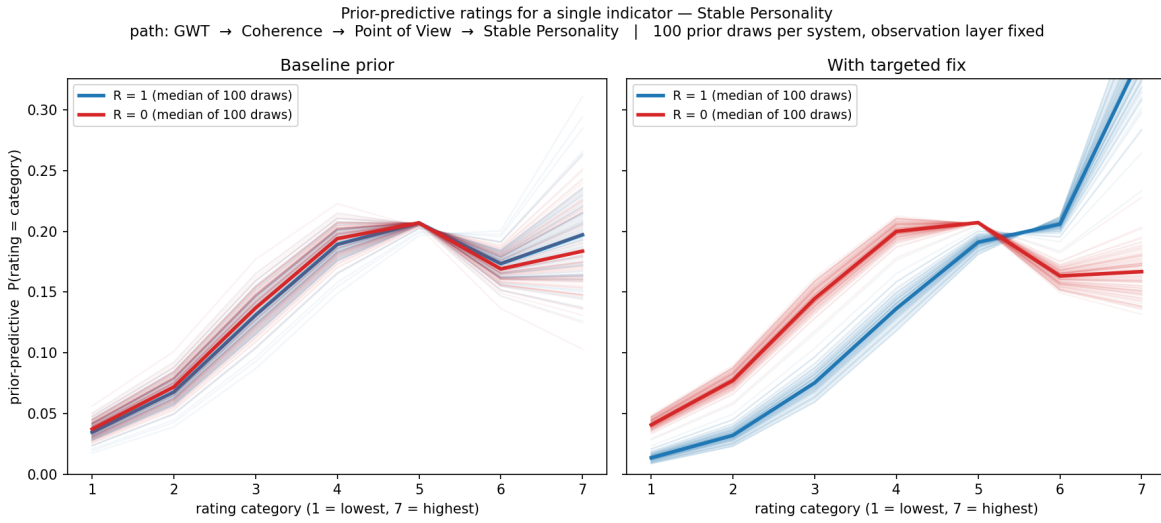


Figure 2: Prior-predictive ratings for the indicator Stable Personality, 100 prior draws per variant per root state

Read off:

- **Baseline (left).** The median rating distributions for $R = 1$ and $R = 0$ are essentially overlaid at every category. Underlying q_j medians are 0.33 ($R = 1$) vs 0.30 ($R = 0$), a gap of 0.03. For this indicator under the original prior, knowing the system is conscious tells us almost nothing about the rating the model predicts.
- **With the targeted fix (right).** The medians separate visibly. q_j medians are 0.65 ($R = 1$) vs 0.26 ($R = 0$), a gap of 0.39. There's still spread in the 100-draw spaghetti

and some overlap, but the central tendency now meaningfully distinguishes the two root states.

Same story as the all-indicator-averaged plot above, but at a named leaf: the prior was preventing the consciousness root from saying anything useful at the rating layer for this specific indicator.

Supporting — q_j vs tree depth

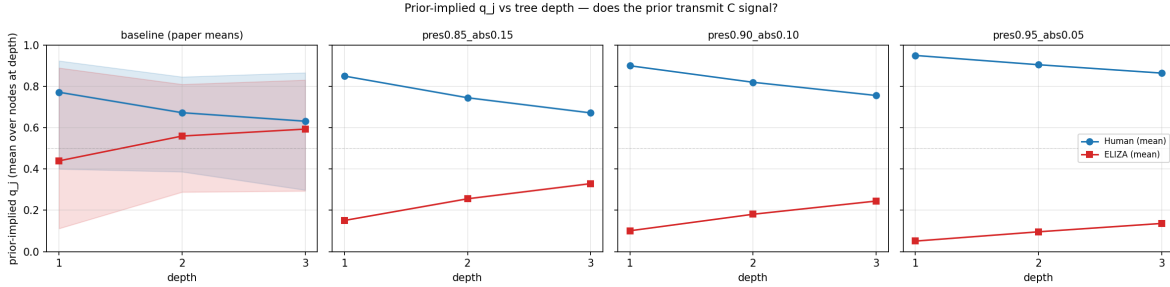


Figure 3: Prior-implied q_j vs tree depth, baseline vs three asymmetric prior overrides

The same story one level upstream of the ratings. q_j is the prior-implied probability that indicator j is in the present state, given root state R .

- **Baseline (leftmost panel).** Human (blue) and ELIZA (red) curves converge by depth 3 — root signal collapses with depth. The shaded bands show the spread of q_j across nodes at each depth (different (support, demandingness) labels give different prior β s).
- **Override panels.** With $\beta_{\text{pres}}, \beta_{\text{abs}}$ pushed away from each other to fixed values across all nodes, the curves stay separated at every depth. Bands are absent here only because the override sets every node's β to the same value, so q_j has no spread within a depth — not because the override is somehow more certain.

My fix — targeted, not a blanket prior swap

- **Targeted, not global.** I override only specific lower-edge nodes (`TARGETED_OVERRIDE_NODE_KEYS`) where the prior was bottlenecking transmission, rather than rewriting every $\beta_{\text{pres}}, \beta_{\text{abs}}$ in the tree. Each targeted edge gets a logit-Normal prior centred near 1 (presence) or 0 (absence), so $\beta_{\text{pres}} - \beta_{\text{abs}}$ stays large enough at those edges to transmit signal.
- **Why targeted matters.** The support / demandingness mapping that produces the existing β priors is informed by domain experts, per stance — it's a substantial body of work and I didn't want to override it wholesale just to fix this leak. The targeted overrides are deliberately conservative.

- **Honest take.** This is the minimum-invasive fix that lets recovery work on synthetic data; it isn't the right end-state. The prior-predictive plot above shows the leaf-level behaviour is still poor, and the deeper question — *whether the support / demandingness $\rightarrow \beta$ translation table itself produces sensible joint prior-predictive behaviour* — is what I'd want to revisit collaboratively. See the follow-up cell.

Recovery — 10-seed synthetic check (brief)

Including this as a sanity check rather than headline material; will likely skim or skip in the talk. A 10-seed synthetic recovery pilot under the targeted-fix variant passes the basic checks: posterior $P(R = 1 \mid \text{data})$ for the unanchored systems tracks the true root state cleanly when the synthetic data carries signal (Chicken: 9/10 correct calls; LLMs: 5/5 correct on truly-absent, 2/5 correct on truly-present where the data carries the signal). Sampler clean across all 10 seeds: no divergences, $\hat{R} \approx 1$.

Per-seed dot plot and CSV are alongside this notebook (`rho_recovery_dotplot.png`, `rho_recovery_per_seed.csv`) for anyone who wants the detail. Caveats around what posterior SD on C (vs ρ) actually tells us are noted in Issue 2 below.

Follow-up — revisit support / demandingness $\rightarrow \beta$ translation, collaboratively

This is the most important next step on the prior side, and I want to flag it clearly so it doesn't read as me sidelining the existing work.

The targeted fix unblocks transmission, but it patches the symptom. The deeper question is whether the **support / demandingness $\rightarrow \text{Beta}(\alpha, \beta)$ translation table** itself produces sensible joint prior-predictive behaviour. Each individual prior makes domain sense in isolation; the product across edges is what was off, and the prior-predictive at the leaves (above) shows even the targeted fix isn't where it should be.

A general lesson worth flagging here, since I think it bites for elicitation in this kind of model: it's natural to set a prior for one variable at a time so that the marginal looks reasonable, while not noticing that the joint behaviour when all the priors are applied together is something quite different from what was intended. Issue 1 is a clean example. Each β prior makes domain sense per edge, but the product across the path collapses the C signal in a way no single edge looks responsible for. Catching this needs joint prior-predictive checks at the leaf level (the figure earlier in this section), not edge-level inspection.

I'd like to keep the existing labels and the existing support / demandingness elicitation intact, and just revisit how that table maps to Beta priors so that the joint prior-predictive at the rating level matches expert intuition. Ideally with Matilda, given her ownership of this layer.

A minimal version that could land in the remaining SPAR window (rather than a full re-elicitation):

- Keep all existing labels fixed.
- Reparameterise the table so demandingness controls a baseline log-odds and support controls an odds-lift on top of it: roughly, $\text{logit}(\beta_{s,d}^{\text{abs}}) = a_d$ and $\text{logit}(\beta_{s,d}^{\text{pres}}) = a_d + \lambda_s$ with $\lambda_s > 0$. This separates “what the conditional means” from “how confident we are in the translation”.
- Elicit only a few anchor judgements: plausible $P(z_u = 1 \mid z_v = 0)$ for low / medium / high demandingness, plausible odds-lift for weak / medium / strong support.
- Tune the table against simple prior-predictive targets at the rating level (high mass on cat 7 under $R = 1$, on cat 1 under $R = 0$) plus reasonable depth-wise q_j gaps.
- Land it as a sensitivity branch alongside the targeted fix, not as a replacement, until you (Matilda) sign off.

One related modelling idea worth flagging. β^{pres} and β^{abs} at the same edge are likely to be correlated in expert intuition: an indicator that strongly signals “parent present” should also be unlikely to fire when the parent is absent, so high β^{pres} tends to go with low β^{abs} . The current model treats them as independent given the (s, d) labels. The reparameterisation above couples them implicitly through the shared baseline a_d , which is one way to encode that correlation. There may also be more explicit options (a joint prior over $(\beta^{\text{pres}}, \beta^{\text{abs}})$ per (s, d) with a negative correlation parameter). Worth thinking about explicitly when revisiting the table.

Process reflection — Bayesian workflow lens

Reflecting on how I ended up here: the prior-predictive plot above is a quick check that doesn’t need any data, and it would have surfaced Issue 1 directly if I’d run it earlier. Gelman et al. 2020, *Bayesian Workflow* (arXiv:2011.01808), Figure 1, places prior-predictive checks early in the workflow for exactly this reason. They catch prior-side mis-specifications before they get tangled up with inference and data-fit questions.

The first phase of this project, the ordinal observation layer plus the product-sum / belief-propagation marginalisation that lets us run NUTS quickly on a normal laptop, was important groundwork, but it’s largely orthogonal to the tree-prior issue. Once that fast-iteration loop existed, applying more of this workflow systematically (prior-predictive, then fit, then validate computation, then posterior-predictive, then modify, looping back) is what would have caught Issue 1 sooner.

A path forward I’d advocate for: take the figure above seriously as a process and walk through it explicitly. Concretely, that would mean prior-predictive checks against domain expectation before any structural change (Issue 1 follow-up); fake-data simulation with known continuous- C truths before fitting real data (Issue 2 implementation); and simulation-based calibration on the headline quantities (ρ_s , C , key β groups) before the model is treated as ready for the next stage. Less ad-hoc diagnostic chasing, more of the workflow loop.

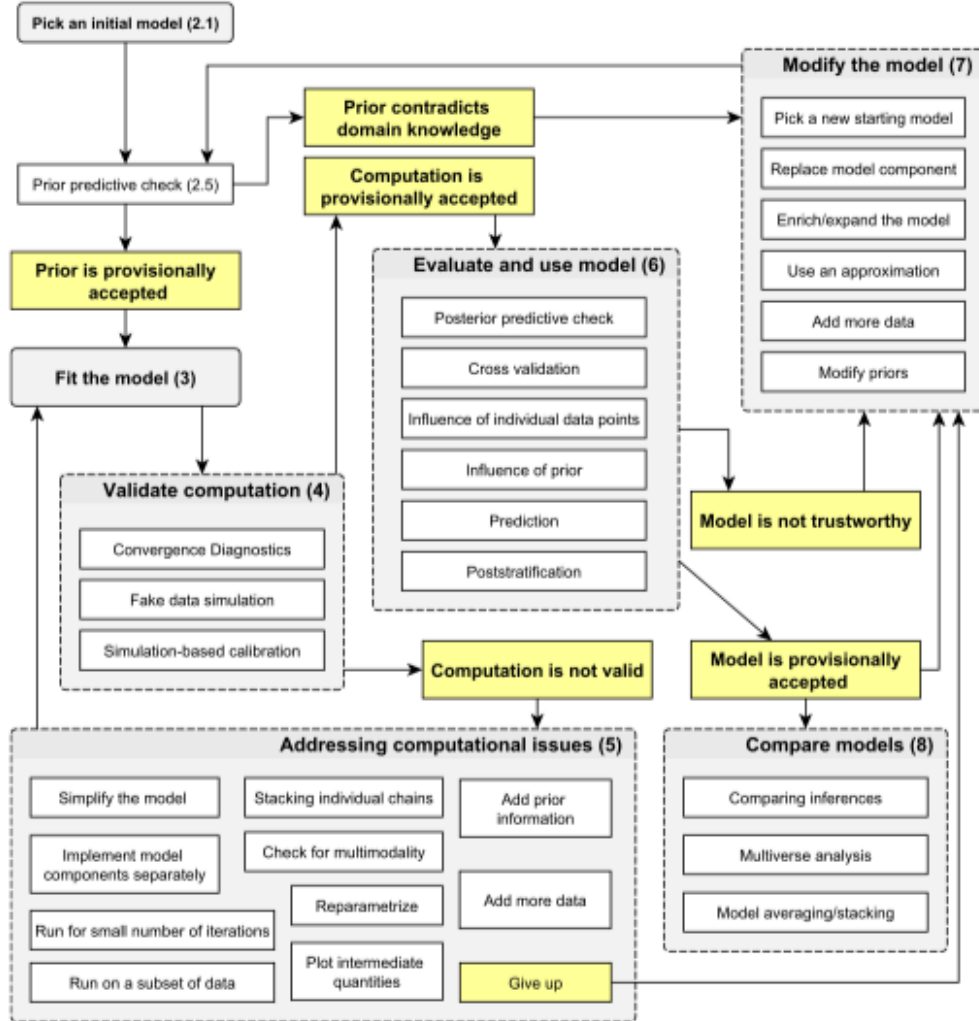


Figure 1: Overview of the steps we currently consider in Bayesian workflow. Numbers in brackets refer to sections of this paper where the steps are discussed. The chart aims to show possible steps and paths an individual analysis may go through, with the understanding that any particular analysis will most likely not involve all of these steps. One of our goals in studying workflow is to understand how these ideas fit together so they can be applied more systematically.

Figure 4: Gelman et al. 2020, Bayesian Workflow, Figure 1 — overview of workflow steps

A personal observation on LLM tools in this kind of work. I’ve found LLM tools useful for code writing, drafting documents like this notebook, and as a partner for idea generation. Where they’ve been less useful, at least with the prompts and harness I’ve been using, is the high-level workflow decisions themselves: when to modify a model, what to compare it against, what priors to set, what to test next. Those calls have still needed to be made by hand. A more carefully designed prompt or scaffold might change that; in practice the workflow loop has been a manual thing for me.

Issue 2 — binary root + extreme references can’t well-identify middle systems

How fixing Issue 1 made this visible

Once root signal actually reaches the indicators, the model effectively has two regimes: $R = 1 \rightarrow$ Human-like ratings, $R = 0 \rightarrow$ ELIZA-like ratings. Middle systems (Chicken, current LLMs) sit empirically between, but the only way the binary root can produce middle-spread behaviour is by mixing those two extreme regimes — so the headline $P(R = 1 \mid \text{data})$ for these systems ends up only weakly informed. The fit to expert ratings can look fine even when the underlying root probability is barely constrained.

A subtle but important point about how to read this: in the current model, each system has only one binary R realisation, drawn from $\text{Bernoulli}(C)$. That means the data can identify the realised R cleanly while still leaving very little to learn about the parameter C itself — C effectively gets only one Bernoulli pseudo-observation. So the “posterior SD on C doesn’t shrink” diagnostic from earlier weeks is partly a feature of the hierarchy, not just a flat-likelihood signal. The right diagnostic is the posterior $P(R = 1 \mid \text{data})$ and the log Bayes factor (above), not the contraction of C .

Visual — real expert rating distributions per system

Generated this morning from `data_cache.json`, pooled across all stances and indicators.

Read off:

- **Human:** 92% at cat 7. **ELIZA:** 99% at cat 1. Reference anchors sit at the two extremes.
- **Chicken:** spread across the upper-middle. 38% at 7, 21% at 5, 15% at 6, 10% at 4, 8% at 3. *Not* a high-low bimodal pattern.
- **LLMs (2024):** most concentrated single category is cat 1 (37%), but every middle category (2–6) carries 7–11% and cat 7 has 21%. Spread broadly across the whole range.

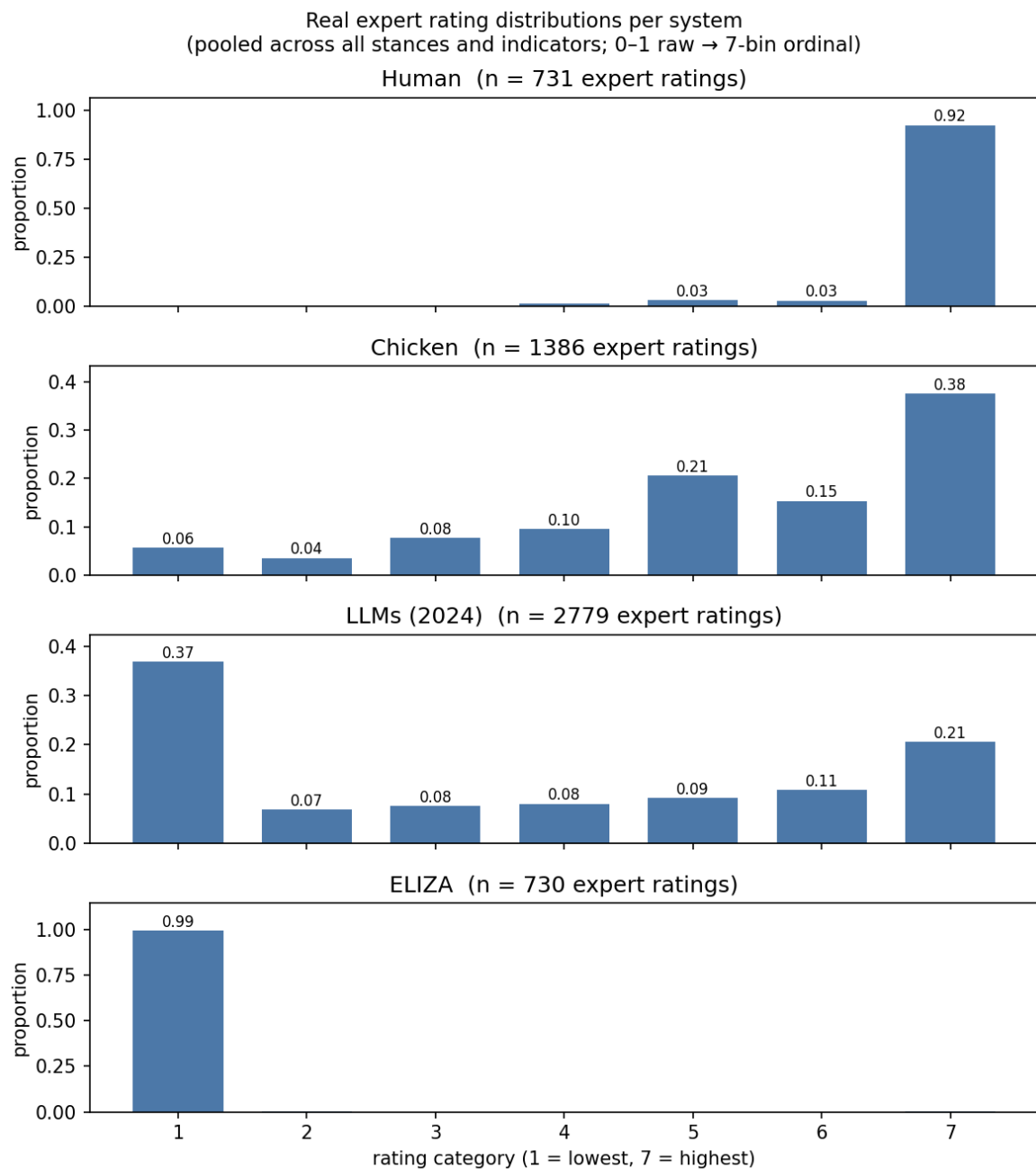


Figure 5: Real expert rating distributions per system, pooled across all stances and indicators

The middle systems sit between the extremes, with substantial mass in middle categories that a binary-root + Human / ELIZA-anchor model can only produce as a probability mixture of the two extreme regimes.

Aside — and a check on the synthetic validation framework

The previously-shared synthetic-validation plot above shows Chicken and LLMs as roughly bimodal at categories 1 and 7. That apparent bimodality is an artefact of pooling across 10 seeds where the synthetic root R was randomly 0 or 1 per seed — when half the seeds have $R=1$ (high ratings) and half have $R=0$ (low ratings), the pooled distribution looks bimodal even though no single synthetic seed produces the spread-across-the-middle distribution we see in real data.

So the synthetic data the validation has been generating doesn't resemble real expert ratings for middle systems. That's another reason synthetic validation results, in isolation, are limited evidence about real-data behaviour.

It's not (mainly) the leaf model

I tested a few alternatives at the leaf: direct- q , mixture, and sweeping the number of raters per indicator. None of them shrank the posterior on C noticeably. **Caveat I should have caught earlier:** under the current binary-root hierarchy, the posterior SD of C is structurally bounded (because C only gets one Bernoulli pseudo-observation from R), so this diagnostic was always partly going to plateau. The right metric is whether the posterior on the realised R (i.e. the ρ above) sharpens. I'd want to re-do the leaf comparison once the root structure is sorted (Issue 2 fix below) before declaring the leaf fully exonerated.

Worth flagging on a different point: some of the posterior-predictive plots I shared in earlier weeks were focused on the rating / ordinal layer, where I was checking that changes I was making to the leaf preserved good behaviour from q_j onwards. Those PPCs looked fine because the rating layer was fine. The issues we now see are upstream of q_j : first in how the tree propagates root information (Issue 1), and now in how the binary root represents systems that sit between Human and ELIZA. I could have signposted earlier which layer those diagnostics were checking.

Suggested next steps (modelling)

Roughly in order of impact and feasibility-this-week.

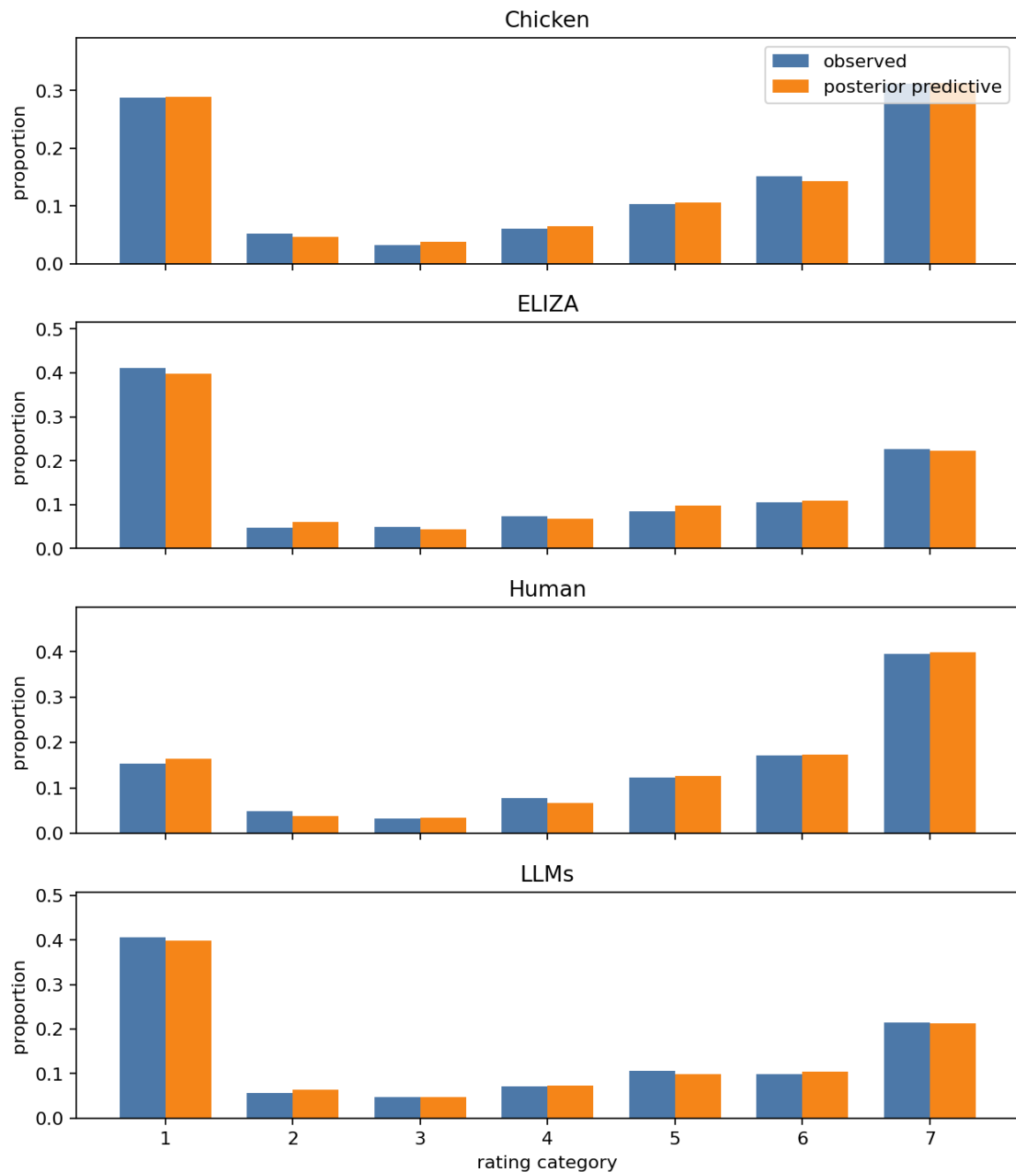


Figure 6: Synthetic validation observed-vs-predicted rating bars per system

1. **Continuous C at the root.** Replace the binary R_s with a continuous root probability $C_s \in [0, 1]$, so middle systems can be fit at intermediate values directly rather than as mixtures of two extreme regimes. Probably the most promising change, and the most actionable in the time remaining: the model spec is well-defined enough that I think a working implementation is feasible within the week with a coding agent + a Pro chat for the maths, if the group thinks this is the right direction. *(Implementation aside: there’s a meaningful choice between (a) renaming C but keeping a single shared $R \mid C \sim \text{Bernoulli}(C)$, which keeps the binary-conscious-or-not semantics but doesn’t actually solve the one-Bernoulli hierarchy issue noted above; and (b) letting C enter the likelihood directly as the per-top-level-child mixing weight $p_u(C) = \beta_u^{abs} + C(\beta_u^{pres} - \beta_u^{abs})$, so C accumulates evidence from multiple top-level features. Option (b) is the principled fix; it does shift the headline from “ $P(R = 1 \mid \text{data})$ ” to “ $E[C \mid \text{data}]$ ”, which is worth flagging as a small but real estimand reframe. We can pick this up when scoping the implementation.)*
2. **Graded reference systems.** Add anchor systems at intermediate root probabilities (chimpanzee toward the upper end is the most defensible candidate I can think of). *Big asterisk:* assigning a justifiable reference probability for anything that isn’t an extreme is hard, and likely stance-dependent. Worth recommending in principle, but the implementation cost is mostly in justifying the numbers, not in the model code.
3. **Cross-stance pooling.** Hierarchical structure across stance variants instead of independent per-stance C_s , borrowing strength across stances for the same system. Most likely beyond what can land before SPAR ends, and relevant to Matilda’s work on tree structure, so worth flagging as a longer-term direction. (Note: this is *not* blocked by the exact-tree marginalisation. Pooling lives in a continuous hyperparameter above the per-stance roots; each stance’s tree likelihood is still computed via the same belief-propagation pass and multiplied. Earlier I’d flagged this as a tree-shape constraint, but that’s only a constraint on the discrete indicator graph, not on hyperparameters above it.)
4. **Open question: joint shared tree (current) vs per-system tree (original).** Both have known issues. Per-system has historically had per-system parameters absorb rating-distribution differences without informing C . I’m leaning toward sticking with the joint shared tree and trying to make middle systems work via (1) and / or (2), but this isn’t a strong view, and group input would be useful.

Key Questions

1. **Support / demandingness $\rightarrow \beta$ translation revisit.** Each individual prior is domain-informed; the joint prior-predictive at the leaf isn’t where it should be (see

the Issue 1 main visual). I'd like to revisit the translation collaboratively (Matilda especially) — keeping existing labels and elicitation, adjusting only how that table maps into β priors. What's the appetite for this and how would you want to scope it?

2. **Joint shared tree vs per-system tree.** For the final report and for the Issue 2 fixes, do we have a settled preference, or is this still open? My lean is joint shared tree + continuous C , but I'd rather hear the group's read before committing.

(*Optional, only if it comes up*). Is there ongoing data-collection / survey-design work where input from this modelling would be useful? Holding off on specific recommendations until Issue 2 is closer to resolved — recommending more raters or particular structures on the basis of the current model would risk validating decisions against a model we know isn't representing middle systems.

Actions for next week (final SPAR week)

- **Final report.** Main deliverable. Joint with Matilda and Arvo (others as relevant). Happy to contribute substantially to the writing.
- **Implement leading Issue-2 candidate (continuous C).** Feasible within the week if the group endorses it and the spec is clear; happy to prototype this with a coding agent + Pro for the maths.
- **Re-run synthetic validation against the fixed model.** More meaningful once Issue 1 + Issue 2 changes are in (and once the synthetic generator is producing data that resembles real expert ratings for middle systems).
- **Real-data fits with the fixed model,** so we have current per-system C numbers to hand off.
- *Open to delegating poster / video to whoever's keen on those; I'd prefer to spend the available time on modelling and the report.*

Where the code lives

- Latest snapshot: branch `ryan/spar-handover-2026-05-12` on github.com/arvomm/dcm-spar.
- This week's directories of interest: `notebooks/asymmetric_prior_sweep_2026-05-10/`, `notebooks/synthetic_validation_2026-05-06/`, `outputs/main_synthetic_validation/`, `outputs/main_synthetic_validation_pilot_20260511/`. The two plot generators for this notebook live alongside it: `generate_prior_predictive_ratings.py` and `generate_real_observed_ratings.py`.